ECE 285 Final Project Report: *SolarInvest Agent*

Manasvin Surya Balakrishnan Jaikannan
UC San Diego
mbj@ucsd.edu

Shubhan Mital
UC San Diego
smital@ucsd.edu

Sri Mihir Devapi Ungarala
UC San Diego
sungarala@ucsd.edu

Abstract

This project proposes SolarInvestAgent, a novel intelligent decision-support system that optimizes the sizing of residential Photovoltaic (PV) systems and Battery Energy Storage Systems (BESS). Unlike standard Large Language Models (LLMs) that rely primarily on heuristic reasoning, the system integrates a rigorous Mixed-Integer Nonlinear Programming (MINLP) formulation to perform constrained optimization. Following the methodology established by Zhou et al. (2018), the agent explicitly models tariff structures (Stepwise and Time-of-Use), seasonal weather variability, and battery degradation constraints to produce mathematically grounded investment recommendations. The system was evaluated using five years of historical household consumption data to capture demand variability and determine the optimal PV technology, system capacity, and installation size that minimize total lifecycle cost under tariff, weather, and degradation constraints. The agentic system reduces panel-count RMSE by $13.8\times$ and capex MAE by $2.3\times$ relative to the LLM-only baseline, with all recommendations validated for physical and budget feasibility. The agent is implemented as a LangGraph-orchestrated pipeline with tool calls to a HiGHS MILP solver, an Open-Meteo irradiance API, and a retrieval-augmented knowledge base, with a three-stage validation loop ensuring all outputs are physically and financially feasible before delivery. Despite strong performance, this evaluation has several limitations. One key limitation is that, even within San Diego, it cannot be fully adopted as experiments are restricted to single-family residential scenarios. This limits generalizability to other climates, building types, and utility tariff structures.

1 Introduction and Related Work

Residential solar adoption is accelerating, yet sizing a PV-BESS system optimally remains difficult: Time-of-Use tariffs, climate variability, and battery aging interact in ways that defy simple rules of thumb. An incorrectly sized system has real consequences - an oversized installation wastes capital, an undersized one fails to offset grid costs, and a financially inconsistent recommendation erodes user trust in AI-assisted energy planning. Existing workflows are either technically rigorous but inaccessible to non-experts, or LLM-driven but heuristic and unverifiable.

Given inputs - location, roof dimensions, household budget, tariff plan, EV count, and occupancy - the system must output an installable PV panel count, battery option, projected NPV, and payback period that minimize lifecycle electricity cost subject to power-balance, SoC, roof-area, and budget constraints. Success is defined as panel-count RMSE $\leq$20 panels and 100% budget feasibility relative to the deterministic solver reference.

- **Method.** SolarInvestAgent, a hybrid architecture that couples a MILP optimizer with an LLM orchestration layer. The LLM handles natural-language interaction and explanation; all numerical decisions are delegated to a constrained solver, enforcing feasibility by construction.
- **Evaluation.** We compare four configurations - a deterministic HiGHS solver (ground truth), an LLM-only track (Track A), and two agentic modes (Track B1 reasoning, Track B2 non-reasoning) - across 20 benchmark scenarios using panel-count RMSE, capex MAE/MAPE, and budget feasibility.

Preprint.

- **Outcome.** The agentic tracks reduce panel-count RMSE by 13.8× (20.05 vs. 277.35) and capex MAE by 2.3× ($26,902 vs. $60,585) relative to the LLM-only baseline, while improving budget feasibility from 25% to 40%.

Related work. Optimal sizing of residential PV and battery systems has been extensively studied as a constrained optimization problem. Zhou et al. established the foundational MINLP formulation, showing that tariff structure critically determines whether PV-only or combined PV-BESS installations are optimal - stepwise tariffs favor PV-only, while Time-of-Use pricing makes battery storage economically attractive through energy arbitrage Zhou et al. [2018]. Subsequent work extended this to stochastic settings incorporating battery degradation, demand response, and household consumption variability, collectively establishing that accurate modeling of tariffs, load, and solar generation is essential for viable residential sizing Afzal et al. [2018]; Hanna et al. [2017]; Wang et al. [2024].

LLMs have been applied to energy forecasting, grid coordination, and policy analysis Bahi et al. [2026], but operate as advisory systems without feasibility guarantees - a critical gap when incorrect recommendations translate to financial loss or uninstallable configurations. The ReAct paradigm Yao et al. [2022] and Reflexion Shinn et al. [2023] showed that LLMs can ground outputs in external computations through structured reasoning-and-tool-calling loops, with frameworks like LangGraph making this practical in production pipelines.

We combine both lines of research: SolarInvestAgent embeds a constrained MILP solver within a LangGraph agentic loop, using the LLM for orchestration and explanation while delegating all numerical decisions to the solver. Unlike prior LLM energy advisors, feasibility is enforced by construction and validated through a three-stage checking pipeline before any recommendation reaches the user.

2 Problem Formulation

We model a grid-connected residential energy system consisting of a photovoltaic (PV) array, a battery energy storage system (BESS), and grid interaction. The system operates over discrete time steps indexed by $t \in \{1, \ldots, T\}$, where $T = 8760$ represents hourly resolution over one year. For stochastic extensions, scenarios are indexed by $s \in \mathcal{S}$. Decision variables and exogenous notations are summarized in Appendix Table 7.

The objective is to minimize the total electricity cost over the planning horizon:

$$\min C_{\text{total}} = C_{\text{PV}} + C_{\text{BESS}} + C_{\text{O\&M}} + \sum_{t=1}^{T} \left[P_{\text{buy}}^{\text{grid}}(t) \cdot \lambda_{\text{TOU}}^{\text{buy}}(t) - P_{\text{sell}}^{\text{grid}}(t) \cdot \lambda_{\text{NEM}}^{\text{sell}}(t) \right] \tag{1}$$

subject to the following constraints:

Constraints	Equations
(1) Power Balance	$P_{\text{load}}(t) = P_{\text{buy}}^{\text{grid}}(t) - P_{\text{sell}}^{\text{grid}}(t) + P_{\text{PV}}(t) + P_{\text{dch}}(t) - P_{\text{ch}}(t)$
(2) Battery SoC Dynamics	$\text{SoC}(t) = \text{SoC}(t-1) + \eta_{\text{ch}} P_{\text{ch}}(t) - \frac{P_{\text{dch}}(t)}{\eta_{\text{dch}}}$
(3) SoC Bounds	$0.2 \cdot E_{\text{BESS}} \leq \text{SoC}(t) \leq 0.8 \cdot E_{\text{BESS}}$
(4) Charge/Discharge Exclusivity	$x_{\text{ch}}(t) + x_{\text{dch}}(t) \leq 1, \quad x_{\text{ch}}, x_{\text{dch}} \in \{0, 1\}$
(5) PV Generation Constraint	$P_{\text{PV}}(t) = S_{\text{PV}} \cdot \frac{G(t)}{G_{\text{ref}}}$

Table 1: Optimization constraints and their mathematical forms.

Outputs and verification. The model outputs optimal sizing decisions $(S_{\text{PV}}^*, E_{\text{BESS}}^*)$ with associated cost and savings estimates, structured as two recommendations (*optimal* and *recommended*) plus a shared evidence block. A recommendation is accepted only if it passes three validation stages: syntactic (valid JSON), schema (structural correctness and value bounds), and logical (budget feasibility and power-balance consistency at every timestep). Failed checks trigger an automatic repair loop that prompts the model to regenerate output under the same schema constraints, preserving evidence traceability to computed features and retrieved context.

3 Methodology

3.1 System Overview

We design a two-track architecture for residential PV sizing: (i) a direct LLM reasoning pipeline, and (ii) SolarInvestAgent, an agentic optimization-driven system. Both tracks share a common data and feature backbone but differ in how decisions are generated. The separation is motivated by a fundamental mismatch between generative reasoning and constraint satisfaction: LLMs produce coherent explanations but frequently violate physical and numerical feasibility, and increasing prompt complexity leads to context overload rather than improved reliability. In the agentic track, deterministic tools handle constrained computation while the LLM synthesizes and communicates decisions.

3.2 Shared Data and Feature Pipeline

Both tracks share a preprocessing pipeline covering data extraction, stochastic household modeling, feature engineering, and retrieval augmentation.

3.2.1 Data Sources and Extraction

The system integrates three primary data streams: (i) five years of historical weather data (irradiance, temperature, cloud cover), (ii) regional electricity demand data, and (iii) derived household-level consumption profiles. These are temporally aligned and aggregated into weekly representations, forming a structured dataset: The full set of modeled San Diego County locations and their geospatial coordinates is listed in Appendix A.6 (Table 10).

$$\mathcal{D} = \{(w_t, l_t)\}_{t=1}^{T}, \quad T \approx 260$$

3.2.2 Stochastic Household Modeling

To bridge the gap between regional and household-level data, we construct a stochastic generative model producing realistic consumption profiles from geographic and behavioral factors. Household load combines multiplicative climate, demographic, and infrastructure scaling factors with additive EV-charging and solar-offset terms; deterministic seeding ensures reproducibility. The full load-factor definitions and TOU tariff assumptions are provided in Appendix A.2 and Appendix A.7 (Table 11), respectively.

3.2.3 Feature Engineering and Aggregation

Raw time-series data is compressed into over 75 domain-specific features spanning consumption patterns, solar potential, financial metrics, and risk indicators. Key sensitivity features include price-ROI sensitivity (`roi_under_price_change`, evaluated at $\pm 10\%$ electricity price perturbations) and irradiance-ROI sensitivity (`roi_under_irradiance_change`, computed by rescaling weekly irradiance by $\pm\delta$). Volatility-based risk features are derived as the coefficient of variation of weekly load ($\mathrm{CV}_{\mathrm{load}}$) and weekly irradiance ($\mathrm{CV}_{\mathrm{irr}}$), and combined into a unified risk score used for uncertainty-aware ranking. Full definitions of all sensitivity and volatility features, together with the complete feature pipeline, are provided in Appendix A.3 (Figure 5).

3.2.4 Retrieval-Augmented Knowledge

Both pipelines share a retrieval layer that injects domain knowledge on PV panel technologies, battery systems, tariffs, and installation practices into the reasoning context. Panel hardware assumptions - ranging from REC Group (Alpha Pure, \$2.85/W_p, 22.6% efficiency) to Aiko Solar (Neostar 2P, 24.3% efficiency) - are summarized in Appendix A.9 (Table 13).

3.3 LLM-Based Design (Baseline)

The LLM baseline receives structured features, retrieved knowledge, and explicit constraints through a hierarchical prompt and directly produces PV sizing recommendations.

3.3.1 Prompt-Based Reasoning

The prompt is structured into three components: (i) feature-engineered data context, (ii) retrieved knowledge, and (iii) explicit task instructions. The instruction block follows an 8-step workflow covering data review, anomaly identification, EV demand accounting, budget and roof feasibility, brand preference, panel count justification, savings estimation, and risk reporting. Hard constraints (budget, roof area, EV load) and soft constraints (brand preference, daytime occupancy) are stated explicitly. A prescribed output schema - requiring panel count, annual production, cost savings, payback, ROI, and caveats - enforces auditable, decision-ready outputs over free-form text.

3.3.2 Limitations of Direct LLM Inference

Despite structured prompts and explicit constraints, the LLM-based approach exhibited three core limitations: (i) inconsistent satisfaction of hard constraints (budget, roof area); (ii) numerical approximation of financial metrics (ROI, payback) rather than strict computation; and (iii) performance degradation under increased prompt complexity due to context saturation. Fundamentally, the model lacks feasibility guarantees, producing outputs that may appear plausible but violate underlying physical or financial constraints.

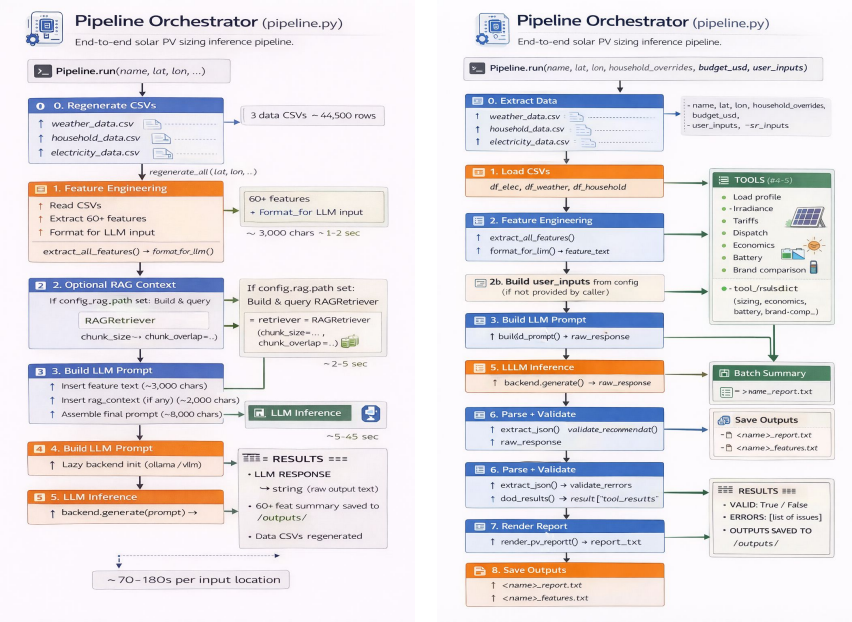


Figure 1: Track A (left) and Track B (right) pipeline overviews used in the system design.

3.4 Agentic Design: SolarInvestAgent

To address these limitations, SolarInvestAgent replaces direct LLM output with a structured planner over computational tools. The LLM reasons and orchestrates tool calls rather than producing final numeric answers directly, enabling correctness through computation while retaining natural-language flexibility.

3.4.1 Tool-Augmented Mathematical Reasoning

All computations are implemented as deterministic functions grounded in physical and financial models. PV generation is derived from irradiance-scaled panel output over an 8760-hour horizon; load and grid interactions are computed in consistent power and energy terms. System sizing and dispatch are jointly optimized via a MILP, with an objective combining annualized capital cost, time-varying import costs, and export credits. NPV and payback are subsequently computed through deterministic cash-flow models, ensuring consistency between technical and economic outputs.

3.4.2 Constraint Enforcement

All feasibility conditions are enforced within the optimization layer. Roof area bounds the maximum panel count (approximately 1.8 m^2 per module), and a budget constraint covers PV, storage, and installation overhead. Operationally, hourly energy balance is enforced at every timestep, battery state-of-charge is bounded between 10% and 100% with round-trip efficiency of 97.5% and charge/discharge rate of 0.5 C, export is capped at available PV generation, and grid imports are non-negative. The full MILP formulation is given in Appendix A.2; hardware cost assumptions are in Table 14.

3.4.3 Agentic Decision Process

The agent operates as a structured planner over a persistent conversation state (full history, no truncation). Each session holds key inputs (location, EV count, occupants, budget, roof, tariff plan) and a pre-computed recommendation JSON; the LLM summarizes this into concise responses (∼1.5–2.5k characters) without rerunning the optimizer. The decision process follows a perception–reasoning–action loop, producing two

configurations - an optimal (efficiency-driven) and a recommended (budget- and area-constrained) solution. Follow-up turns incrementally extend the conversation context without re-invoking the MILP, separating offline deterministic computation from online LLM-driven interaction.

3.5 Implementation Details (Engineering and Inference Optimization)

The system is implemented as a modular LLMOps pipeline with a backend-agnostic inference layer. Production inference uses `grok-4.1-fast` with retry mechanisms, controlled decoding, and strict token budgeting. A lightweight retrieval layer injects domain context into prompts, and a schema validation-and-repair loop ensures outputs conform to predefined formats before being returned to the user.

4 Experiments and Results

4.1 Experimental Setup

Component	Baseline (GT)	LLM Track	Agentic Task 1	Agentic Task 2	Agentic Task 3
Task	Ground-truth reference	Benchmark: Vanilla LLM	Benchmark: Solar-InvestAgent	Grid evaluation 1	Grid evaluation 2
Description	Ground-truth reference for all error metrics	Custom benchmark for stress testing for extreme scenarios	Custom benchmark for stress testing for extreme scenarios	Parameter sweep over uncorrelated features	Parameter sweep over correlated features
Constant parameters	N/A	N/A	N/A	Budget: $25,000 roof length: 8.0 m roof breadth: 6.25 m rate plan: TOU_DR panel brand: auto	EVs: 0 people: 2 daytime occupants: 1 rate plan: TOU_DR
Variable parameters	N/A	N/A	N/A	EVs: 0, 1, 2 people: 1, 2, 3 daytime occupants: 0, 1, 2	roof length: 5, 10, 15 m roof breadth: 5, 10, 15 m budget: $5,000, $10,000, $15,000 panel brand: REC Group, Canadian Solar, Aiko Solar
Locations	La Jolla, Ocean-side	La Jolla, Ocean-side	La Jolla, Ocean-side	La Jolla, Ocean-side, Coronado	La Jolla, Ocean-side, Coronado
Test size	20	20	20	54	314
Models	None	`llama3.1:8b`	`grok-4-1-fast-reasoning`, `grok-4-1-fast-non-reasoning`		
Frameworks	`HiGHS`	Ollama	LangGraph		
Budget	Calls: 20 Tokens: 0 Time: ∼6.3 h	Calls: 20 Tokens:∼55k Time: ∼25 min	Calls: 20 Tokens:∼150k Time: ∼15 min	Calls: 54 Tokens: ∼405k Time: ∼40 min	Calls: 243 Tokens: ∼1.8M Time: ∼3-4 hours
Data splits	N/A (inference only). Each scenario is independent; no training/validation split.				
Common tools	Open-Meteo, EIA load data, SDG&E tariffs, RAG KB				

Table 2: Experimental setup across benchmark and grid evaluation tasks.

4.2 Metrics

All error metrics are computed relative to the **Baseline as ground truth**, over N=20 scenarios each for the *unconstrained* and *budget-constrained* sizing objectives.

Panel count RMSE and MAE.

$$\mathrm{RMSE}_{\text{panels}} = \sqrt{\frac{1}{N}\sum_i \left(\hat{n}_i - n_i^{\mathrm{GT}}\right)^2}, \qquad \mathrm{MAE}_{\text{panels}} = \frac{1}{N}\sum_i \left|\hat{n}_i - n_i^{\mathrm{GT}}\right|.$$

Capex MAE and MAPE.

$$\mathrm{MAE}_{\text{capex}} = \frac{1}{N}\sum_i \left|\hat{C}_i - C_i^{\mathrm{GT}}\right|, \qquad \mathrm{MAPE}_{\text{capex}} = \frac{1}{N}\sum_i \frac{|\hat{C}_i - C_i^{\mathrm{GT}}|}{C_i^{\mathrm{GT}}} \times 100\%.$$

Budget feasibility (higher is better). Per-scenario binary indicator: feasible if predicted CAPEX $\leq$ stated household budget ($\hat{C}_i \leq B_i$). A violation is a hard constraint failure: the proposed system cannot be built within budget.

4.3 Baselines

We evaluate four configurations:

- **Baseline (ground truth).** A deterministic solver-only pipeline that runs the full MILP optimizer directly in Python, without any LLM involvement. Given household inputs, it fetches real irradiance from the Open-Meteo API, models household load from EIA data, selects the optimal panel from a hardware catalog, and solves the constrained PV-BESS sizing problem using `HiGHS` via `highspy`. The output - panel count and gross CAPEX - serves as the reference for all error metrics.
- **Track A (LLM-only).** An LLM-only pipeline that receives structured features and extended retrieval context (panel datasheets, tariff schedules, installation references) but produces recommendations entirely through prompt-based reasoning. No optimization solver is invoked and no hard constraints are explicitly enforced- budget, roof area, and feasibility requirements are stated in the prompt but compliance depends solely on the model's generative behavior.
- **Track B1.** SOLARINVESTAGENT (reasoning mode): `grok-4-1-fast-reasoning` orchestrates the MILP optimizer through step-by-step chain-of-thought reasoning before each tool call with an additional scenario-consistency validation after the optimizer returns results.
- **Track B2.** SOLARINVESTAGENT (non-reasoning mode): `grok-4-1-fast-non-reasoning` orchestrates the MILP optimizer through step-by-step chain-of-thought reasoning before each tool call with an additional scenario-consistency validation after the optimizer returns results.

Fairness. All tracks receive identical input features, tariff schedules, weather data, and panel cost assumptions. The only axes of variation are tool access and reasoning strategy.

4.4 Results

4.4.1 Unconstrained Sizing (Optimal)

Table 3 reports deviation from the Baseline reference under the unconstrained sizing objective, where the goal is pure lifecycle cost minimisation with no enforced budget cap.

Table 3: All tracks vs Baseline reference **Unconstrained sizing** (N=20). Lower RMSE/MAE/MAPE and higher feasibility are better. Baseline feasibility: 65.0% (13/20, 7 violations). Tracks B1 and B2 produce identical numeric outputs on this benchmark; the reasoning mode (Track B1) and non-reasoning mode (Track B2) differ in inference strategy but converge to the same MILP solution. High MAPE values arise because several Baseline capex entries are very small ($127-$760), amplifying percentage errors.

Track	RMSE (panels)	MAE (panels)	Capex MAE ($)	Capex MAPE (%)	Feasibility	Violations
Track A	277.35	186.85	60,585	10,468.5	25.0%	15/20
Track B1	**20.05**	**14.10**	**26,902**	**1,365.2**	**40.0%**	**12/20**
Track B2	20.05	14.10	26,902	1,365.2	40.0%	12/20

Track A vs Tracks B1/B2 Unconstrained. A one-line unconstrained head-to-head summary comparing Track A with Tracks B1/B2 (relative to the Baseline, with ✓denoting the better metric) is provided in Appendix A.8, Table 12. Tracks B1/B2 win on all dimensions: panel count RMSE is 13.8× lower (20.05 vs. 277.35), capex MAE is 2.3× lower ($26,902 vs. $60,585), and budget feasibility is higher (40% vs. 25%), demonstrating that MILP-grounded recommendations align substantially more closely with the deterministic Baseline.

4.4.2 Budget-Constrained Sizing (Recommended)

Table 4 reports deviations from the Baseline reference under the budget-constrained sizing objective, where all methods receive an explicit household budget to respect.

Track A vs Tracks B1/B2 Budget-constrained. The detailed head-to-head under the budget-constrained objective is provided in Appendix A.5 (Table 9). As noted in the table note, Track A's feasibility advantage stems from over-estimation: larger recommended panel counts incidentally correspond to capex figures that remain within budget in more scenarios, and this does not reflect superior constraint enforcement. As shown in Appendix Table 9, Tracks B1/B2 achieve lower error across all accuracy metrics: panel RMSE is 2.3× lower (37.80 vs. 86.44), capex MAE is 2.4× lower ($7,670 vs. $18,220), and capex MAPE is 2.4× lower (732% vs. 1,738%).

Table 4: All tracks vs Baseline reference **Budget-constrained sizing** (N=20). Baseline feasibility: 100.0% (20/20, 0 violations). Tracks B1 and B2 produce identical outputs. Track A has fewer budget violations (3 vs. 5) because its tendency to over-estimate panel counts happens to yield capex figures that stay within budget in more scenarios, not because of tighter constraint enforcement.

Track	RMSE (panels)	MAE (panels)	Capex MAE ($)	Capex MAPE (%)	Feasibility	Violations
Track A	86.44	63.70	18,220	1,738.2	85.0%	3/20
Track B1	**37.80**	**26.00**	**7,670**	**732.2**	75.0%	5/20
Track B2	37.80	26.00	7,670	732.2	75.0%	5/20

4.4.3 Sensitivity and Robustness Analysis of Agentic PV Sizing

Across both grid evaluations, the agent exhibits stable monotonic behavior with respect to supply- and demand-side inputs. In the 243-point sweep (roof dimensions, budget, panel brand), panel count scales near-linearly with roof area; budget acts as a soft constraint that becomes binding only after spatial limits are reached. CAPEX tracks panel count proportionally, with brand-driven offsets of 15-25% for premium selections. In the 72-point sweep (EVs, occupants, daytime presence), panel recommendations are driven primarily by EV count - each additional vehicle producing a step increase - with occupancy contributing a smaller but consistent effect. Location had negligible variance impact across both sweeps, consistent with the similar solar profiles across the San Diego region. Detailed sweep plots are provided in Appendix A.4 (Figures 6-7).

4.4.4 Overall Per-Track Summary

Table 5 consolidates average outputs and errors for all four tracks across both sizing objectives.

Table 5: Per-track summary over 20 scenarios. RMSE and Capex MAE are measured against the Baseline reference (zero by definition for Baseline). Tracks B1 and B2 produce identical numeric outputs; the reasoning (B1) and non-reasoning (B2) inference modes converge to the same MILP solution, indicating that the chain-of-thought step does not alter the numeric recommendation on this benchmark.

Objective	Track	Avg panels	Avg capex ($)	Panels RMSE	Capex MAE ($)	Feas.	Viol.
Unconstrained (Optimal)	Baseline	33.9	21,811	0.00	0	65.0%	7/20
	Track A	220.7	81,602	277.35	60,585	25.0%	15/20
	Track B1	33.6	44,436	**20.05**	**26,902**	**40.0%**	**12/20**
	Track B2	33.6	44,436	20.05	26,902	40.0%	12/20
Budget-constrained (Recommended)	Baseline	32.4	17,394	0.00	0	**100.0%**	0/20
	Track A	84.1	34,812	86.44	18,220	85.0%	3/20
	Track B1	9.9	24,944	**37.80**	**7,670**	75.0%	5/20
	Track B2	9.9	24,944	37.80	7,670	75.0%	5/20

1. *Tracks B1 and B2 are closest to the Baseline reference.* Across both objectives, Tracks B1 and B2 achieve the lowest panel count RMSE (20.05 unconstrained, 37.80 budget-constrained) and lowest capex MAE ($26,902 and $7,670 respectively), confirming that MILP-grounded recommendations align more closely with the deterministic Baseline than LLM-only approaches.
2. *Track A diverges substantially from the Baseline.* Without hard constraint enforcement, the LLM over-estimates system size dramatically: average panels jump from 33.9 (Baseline) to 220.7 (Track A) under the unconstrained objective, and from 32.4 to 84.1 under the budget-constrained objective. This inflates capex MAE to $60,585 and $18,220 respectively 2.3× and 2.4× worse than Tracks B1/B2.
3. *Tracks B1/B2 outperform Track A on all accuracy metrics.* In both objectives, Tracks B1 and B2 achieve lower RMSE, MAE, and MAPE than Track A. Track A records slightly fewer budget violations under the budget-constrained objective (3 vs. 5), but this is a coincidental consequence of its over-estimation pattern rather than improved constraint enforcement.
4. *Reasoning mode does not change numeric outputs.* Tracks B1 (reasoning) and B2 (non-reasoning) produce identical panel counts and capex on all 20 scenarios, indicating that the chain-of-thought step does not alter the MILP solution on this benchmark. The value of the reasoning mode may lie in explanation quality and traceability, which are not captured by these metrics.
5. *Budget feasibility is objective-dependent.* The Baseline achieves 100% feasibility under the budget-constrained objective by construction, but drops to 65% (7 violations) under the unconstrained objective

where no budget is explicitly provided. Tracks B1/B2 incur 5 violations under the budget-constrained objective, all concentrated in scenarios with budgets $\leq$\$9,000 where the MILP's minimum viable configuration (\$12,650) inherently exceeds the stated budget.

4.4.5 Example Trajectory

User: I live in La Jolla. Budget \$15,000, roof 3 m $\times$ 2.5 m, no EVs, 1 occupant, nobody home during the day. What solar system do you recommend?

Agent plan: (1) Fetch La Jolla irradiance via Open-Meteo. (2) Pull single-occupant EIA load profile. (3) Retrieve SDG&E TOU-DR tariff schedule. (4) Call `pv_bess_optimizer`(budget=\$15,000, roof=7.5 m^2, evs=0). (5) Validate JSON output against schema; repair if needed.

Tool call: `pv_bess_optimizer`({budget: 15000, roof_area: 7.5, evs: 0, people: 1, tariff: "TOU_DR"})

Tool output: {panels: 1, capex: \$14,198.80, annual_savings: \$892, payback_yr: 15.9, feasible: true}

Validation: Schema OK; Capex \$14,199 $\leq$ budget \$15,000; SoC bounds [10%-100%] satisfied; Power balance feasible over 8,760 h

Final output: "For your La Jolla home we recommend a **1-panel starter system with BESS** at a total installed cost of \$14,199 - within your \$15,000 budget. Under TOU-DR you can expect \$892/year in savings with a payback period of $\approx$16 years. Given your small roof (7.5 m^2) and low single-person load, one panel with battery storage captures most of the available arbitrage value. Adding a second panel would require a larger budget."

Figure 2: SolarInvestAgent (Track B) trajectory for scenario 6 (*tiny_roof_minimal_load*, La Jolla, Recommended case). The agent calls the MILP optimizer, validates all constraints, and produces a grounded natural-language recommendation. The Baseline (GT) output for the same scenario: 2 panels, capex \$12,903 - within budget but without constraint verification.

5 Conclusion, Limitations, and Future Work

We presented SolarInvestAgent, a hybrid decision-support system that combines LLM-based orchestration with solver-backed optimization for residential PV-BESS sizing. Our results show that embedding a deterministic MILP core substantially improves numerical accuracy, constraint satisfaction, and financial consistency compared to LLM-only baselines. In particular, the agentic approach achieves significantly lower sizing and cost errors while avoiding the infeasible and over-estimated configurations commonly produced by direct LLM inference. More broadly, this work highlights the importance of separating reasoning from computation in LLM systems: while LLMs are effective for synthesis and interaction, reliable decision-making in structured domains requires explicit optimization and verification.

Limitations include fixed annual PV degradation rates, a single-year MILP solved once at query time (missing non-linear aging, tariff escalation, and load growth), and evaluation restricted to single-family residential scenarios in San Diego, limiting generalizability to other climates, building types, and utility structures. Additionally, the current system does not support real-time tariff updates or automated incentive tracking, requiring manual parameter updates as utility rates change.

Future work includes (i) incorporating non-linear degradation curves and automated scraping of live TOU rates and incentives; (ii) replacing the single-year MILP with a multi-stage stochastic program that re-optimizes annually over a 10-25 year horizon with dynamic hardware upgrade planning; and (iii) generalizing to apartment complexes and small commercial buildings with shared battery dispatch, demand-charge tariffs, and community solar allocation; (iv) expanding the framework to additional deregulated energy markets such as Texas (ERCOT), where real-time pricing and distinct grid interconnection policies introduce new optimization challenges; and (v) replacing the single deterministic solver with a multi-agent reinforcement learning architecture in which autonomous agents representing individual households, the grid operator, and energy retailers negotiate dispatch schedules and adapt to non-stationary price signals in a shared environment.

Table 6: SolarInvest Agent: Traceability and Scenario Consistency

Model / Variant	NPV Err. (%)	Payback Err. (yr)	Traceable (%)	Scenario Cons. (%)	Safe Fail (%)	Tool Calls
Vanilla LLM	30	3	80	75	98	0
SolarInvest Agent	1.5	0.2	88	82	97	9

6 Team Member Contributions

Shubhan Mital: I contributed to the problem formulation, including the mathematical equation setup for the optimization framework. I designed and implemented the HiGHS-based baseline solver, subsequently integrating the baseline into the agent-driven framework. I also developed the code pipeline for electricity consumption data processing. In addition, I actively co-developed the project presentation and report, contributing equally with my teammates in creating slides and writing technical content. I developed code to systematically compare the outputs of the baseline solver, LLM-based approach, and agent-based system. Additionally, I ideated and contributed to the architectural design of both the LLM and agent components and provided support in their implementation, particularly in aligning them with the overall optimization pipeline.

Manasvin Surya Balakrishnan Jaikannan: I developed the data schema and built the end-to-end LLM and agentic system. This included designing the agent workflow, integrating optimization tools into the agent's decision layer, and setting up inference pipelines for both Track A and Track B, along with evaluation on a custom benchmark dataset. I also ran parameter sweep experiments to study how variations in inputs affected model outputs and system decisions. On the modeling side, I iterated on system prompts, RAG prompts, and follow-up prompting strategies to improve grounding, consistency, and response quality across multi-turn interactions. I built the SolarInvest Agent chatbot using Gradio, wiring it to Grok models for low-latency inference and handling stateful, context-aware follow-ups. The interface was designed to support structured inputs, tool-triggered reasoning, and seamless user interaction with the agent. In parallel, I contributed to drafting and refining the report and presentation, focusing on clearly communicating system design choices, experiments, and key results.

Sri Mihir Devapi Ungarala: I was responsible for dataset collection, ensuring the project had reliable data for modeling and evaluation. I also contributed significantly to presentation development and report editing, improving both clarity and overall communication quality.

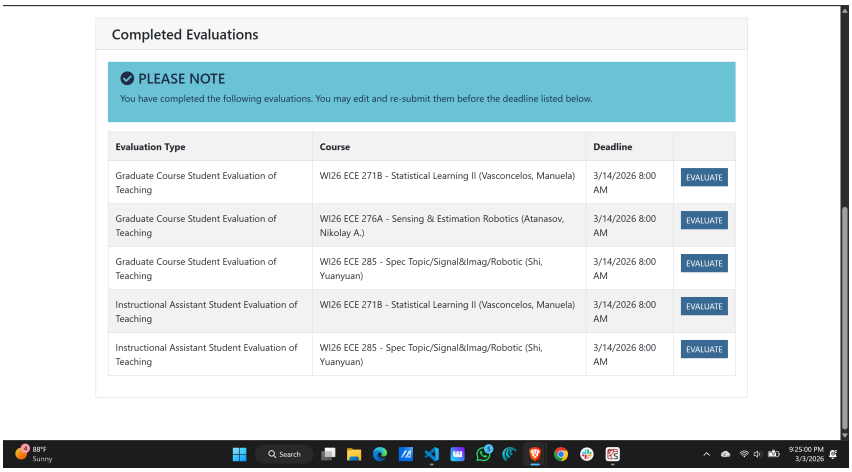
7 Acknowledgments

We thank **Professor Yuanyuan Shi**, **Co-Instructor Daniela Rojas**, and the ECE 285 teaching assistant for their guidance, feedback, and support throughout this project.

We also acknowledge the use of AI-assisted development tools in building this system. ChatGPT (OpenAI) and Claude (Anthropic) were used for code development, debugging, and report drafting assistance. Claude Code and Cursor were used as AI-powered coding environments that accelerated implementation and iteration across the pipeline. OpenAI PRISM and Google Gemini were used for report editing and image generation, including system architecture diagrams and pipeline visualizations. All idea development, technical contributions, experimental design, and conclusions remain solely those of the authors.

8 Bonus Credit: Course Evaluation Submission

We have submitted the course evaluation for ECE 285. Screenshots of submission are provided below.



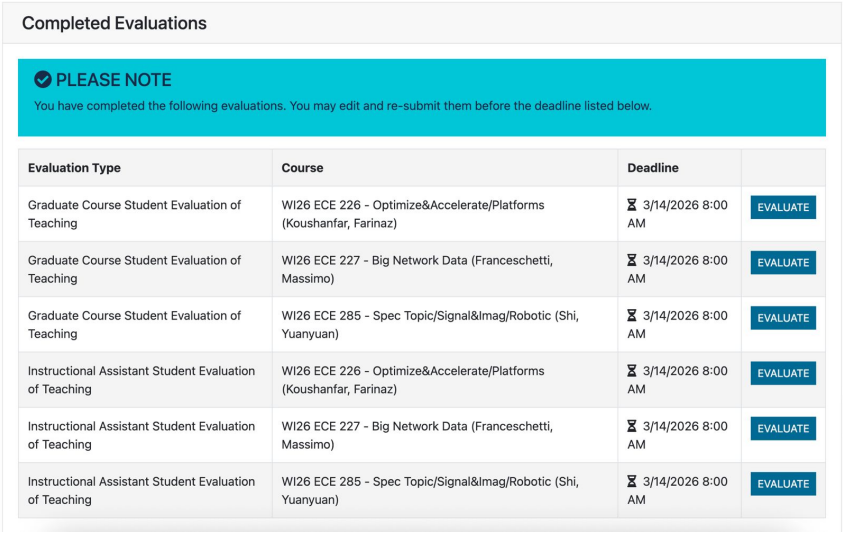
Completed Evaluations

PLEASE NOTE

You have completed the following evaluations. You may edit and re-submit them before the deadline listed below.

Evaluation Type	Course	Deadline	
Graduate Course Student Evaluation of Teaching	WI26 ECE 271B - Statistical Learning II (Vasconcelos, Manuela)	3/14/2026 8:00 AM	EVALUATE
Graduate Course Student Evaluation of Teaching	WI26 ECE 276A - Sensing & Estimation Robotics (Atanasov, Nikolay A.)	3/14/2026 8:00 AM	EVALUATE
Graduate Course Student Evaluation of Teaching	WI26 ECE 285 - Spec Topic/Signal&Imag/Robotic (Shi, Yuanyuan)	3/14/2026 8:00 AM	EVALUATE
Instructional Assistant Student Evaluation of Teaching	WI26 ECE 271B - Statistical Learning II (Vasconcelos, Manuela)	3/14/2026 8:00 AM	EVALUATE
Instructional Assistant Student Evaluation of Teaching	WI26 ECE 285 - Spec Topic/Signal&Imag/Robotic (Shi, Yuanyuan)	3/14/2026 8:00 AM	EVALUATE

Figure 3: Shubhan Mital



Completed Evaluations

PLEASE NOTE

You have completed the following evaluations. You may edit and re-submit them before the deadline listed below.

Evaluation Type	Course	Deadline	
Graduate Course Student Evaluation of Teaching	WI26 ECE 226 - Optimize&Accelerate/Platforms (Koushanfar, Farinaz)	3/14/2026 8:00 AM	EVALUATE
Graduate Course Student Evaluation of Teaching	WI26 ECE 227 - Big Network Data (Franceschetti, Massimo)	3/14/2026 8:00 AM	EVALUATE
Graduate Course Student Evaluation of Teaching	WI26 ECE 285 - Spec Topic/Signal&Imag/Robotic (Shi, Yuanyuan)	3/14/2026 8:00 AM	EVALUATE
Instructional Assistant Student Evaluation of Teaching	WI26 ECE 226 - Optimize&Accelerate/Platforms (Koushanfar, Farinaz)	3/14/2026 8:00 AM	EVALUATE
Instructional Assistant Student Evaluation of Teaching	WI26 ECE 227 - Big Network Data (Franceschetti, Massimo)	3/14/2026 8:00 AM	EVALUATE
Instructional Assistant Student Evaluation of Teaching	WI26 ECE 285 - Spec Topic/Signal&Imag/Robotic (Shi, Yuanyuan)	3/14/2026 8:00 AM	EVALUATE

Figure 4: Manasvin Surya Balakrishnan Jaikannan

References

M. Afzal, Q. Huang, W. Amin, K. Umer, A. Raza, and M. Naeem. Optimal sizing and energy scheduling of photovoltaic-battery systems under different tariff structures. *Renewable Energy*, 129:513–526, 2018. doi: 10.1016/j.renene.2018.06.016.

Ahmed Bahi et al. Large language models for smart grid decision support: A survey, 2026.

California Public Utilities Commission. Net energy metering (NEM) 3.0 net billing tariff decision. Technical Report Decision 22-12-045, CPUC, 2022. URL https://www.cpuc.ca.gov/nem.

Digiqt. Voice agents in solar power: Proven, powerful wins. https://digiqt.com/blog/voice-agents-in-solar-power/, 2025. Accessed: 2026-02-06.

Y. Dong et al. Energyllm: Benchmarking safety and reliability of llms for energy systems. *arXiv preprint arXiv:2403.00000*, 2024.

EnergySage. Solar panel brands: Buyer's guide, 2025. URL https://www.energysage.com/solar-panels/.

R. Hanna, J. Kleissl, A. Nottrott, and M. Ferry. Stochastic optimal battery storage sizing and scheduling in home energy management systems equipped with solar photovoltaic panels. *Energy and Buildings*, 152:290–300, 2017. doi: 10.1016/j.enbuild.2017.07.043.

Q. Hassan, M. Jaszczur, E. Przenzak, and J. Abdulateef. Optimal sizing of rooftop PV and battery storage for grid-connected houses considering flat and time-of-use electricity rates. *Energies*, 14(12):3520, 2021. doi: 10.3390/en14123520.

Dirk C. Jordan and Sarah R. Kurtz. Photovoltaic degradation rates—an analytical review. *Progress in Photovoltaics: Research and Applications*, 21(1):12–29, 2013. doi: 10.1002/pip.1182.

National Renewable Energy Laboratory. Annual technology baseline (ATB): Utility-scale battery storage. Technical report, NREL, 2024a. URL https://atb.nrel.gov.

National Renewable Energy Laboratory. Operations and maintenance costs for solar PV systems. Technical report, NREL, 2024b. URL https://www.nrel.gov/solar.

National Renewable Energy Laboratory. PVWatts calculator, 2024c. URL https://pvwatts.nrel.gov. Accessed: 2026-02.

OnixNet. Introducing the AI-powered solar assistant for energy companies. https://www.onixnet.com/blog/introducing-the-ai-powered-solar-assistant-for-energy-companies/, 2025. Accessed: 2026-02-06.

San Diego Gas & Electric. TOU-DR, TOU-DR1, and TOU-DR2 Residential Time-of-Use Rate Schedules, 2021–2025, 2025. Rates manually extracted from annual regulatory PDF tariff schedules, segmented by effective date. Accessed February 2026.

Noah Shinn, Federico Cassano, Emmanuel Labash, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *arXiv preprint arXiv:2303.11366*, 2023.

X. Wang, Y. Chen, and Z. Li. A stochastic method for behind-the-meter PV-battery energy storage system sizing. *Journal of Energy Storage*, 73:108956, 2024. doi: 10.1016/j.est.2023.108956.

Shunyu Yao et al. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.

Y. Zhou, J. Zhang, X. Yu, D. Zhao, and H. Yan. Optimal sizing of PV and BESS for a smart household considering different price mechanisms. *IEEE Access*, 6:41050–41059, 2018. doi: 10.1109/ACCESS.2018.2858084.

A Additional Figures and Tables

A.1 Notation Summary

Decision variables and exogenous notations are summarized in Appendix Table 7.

Symbol	**Definition**
S_{PV}	Installed PV capacity (kWp)
E_{BESS}	Battery energy capacity (kWh)
$P_{\text{buy}}^{\text{grid}}(t),\ P_{\text{sell}}^{\text{grid}}(t)$	Grid import/export power (kW)
$P_{\text{ch}}(t),\ P_{\text{dch}}(t)$	Battery charge/discharge power (kW)
$\text{SoC}(t)$	Battery state-of-charge (kWh)
$P_{\text{load}}(t)$	Household demand (kW)
$P_{\text{PV}}(t)$	PV generation (kW)
$\lambda_{\text{TOU}}^{\text{buy}}(t),\ \lambda_{\text{NEM}}^{\text{sell}}(t)$	Time-varying buy/sell tariffs ($/kWh)
$\eta_{\text{ch}},\ \eta_{\text{dch}}$	Battery charge/discharge efficiency
$x_{\text{ch}}(t),\ x_{\text{dch}}(t)$	Binary charge/discharge mode indicators
$G(t),\ G_{\text{ref}}$	Hourly irradiance; reference irradiance ($\text{kWh}\,\text{m}^{-2}$)

Table 7: Notation summary for decision variables and exogenous inputs.

A.2 Full Problem Formulation

Symbol	Definition
Irradiance and PV	
G	Annual irradiance (kWh m^{-2} yr^{-1})
$\bar{R}_d$	Daily shortwave radiation (MJ m^{-2} day^{-1})
η_{PR}	PV performance ratio (0.80)
$\phi(h, d)$	Normalised hourly irradiance shape factor
$S(d),\ r$	Sunset/sunrise hour for day d
$p,\ P_{\text{W}}$	Panel count (integer); rated panel power (W)
$\eta_j,\ c_j$	Efficiency and cost of panel type j
Load profile scaling factors	
f_1	Inland/coastal longitude factor
f_2	Latitude-based temperature factor
f_3	Combined inland heat and elevation factor
f_4	Household size factor (n_p = number of people)
f_5	Urban proximity factor (d_c = distance from city centre)
f_6	Coastal/urban adjustment factor
f_9	Stochastic micro-grid variability factor
n_{EV}	Number of electric vehicles
n_{day}	Number of daytime occupants
Battery dispatch	
C	Usable battery capacity (kWh)
$P_c^{\max},\ P_d^{\max}$	Max charge/discharge power (kW)
$c_t,\ d_t$	Charge/discharge power at hour t (kW)
$e_t,\ g_t$	Export to grid / import from grid at hour t (kW)
net_t	Net load: $L_t - P_t^{\text{PV}}$ (kW)
Economics	
r_{inst}	Installation overhead rate (0.10)
r_{ITC}	Investment tax credit rate (0.30)
r_d	Discount rate (0.07)
r_{inf}	Utility inflation rate (0.06)
δ	Annual PV degradation rate
F	Daily fixed grid charge ($0.345/day)
AF	Annuity factor
PVM	Present-value multiplier
N	Planning horizon (10 yr)
y_k	Binary battery option selector
$C_k,\ P_k^{c,\max},\ P_k^{d,\max}$	Capacity and power limits for battery option k
String configuration	
$n_s^*,\ n_p$	Optimal series/parallel panel split
V_{mp}	Panel maximum-power voltage (V)
$N_s^{\text{cell}},\ N_p^{\text{cell}}$	Series/parallel cells per panel

Table 8: Extended notation for appendix formulations.

Irradiance Model. Sunset hour for day d:

$$S(d) = 18 + 2\sin\left(\frac{2\pi(d-80)}{365}\right), \quad r = 6 \text{ (sunrise)} \tag{2}$$

Normalised hourly shape factor:

$$\phi(h, d) = \max\left(0,\ \sin\left(\frac{\pi(h-r)}{S(d)-r}\right)\right) \cdot \mathbf{1}[r \le h < S(d)] \tag{3}$$

Calibrated hourly irradiance ($\eta_{\text{PR}} = 0.80$, G = annual kWh m^{-2} yr^{-1}):

$$G_t = \phi(h_t, d_t) \cdot \frac{G\,\eta_{\text{PR}}}{\sum_{t'} \phi(h_{t'}, d_{t'})} \tag{4}$$

API conversion from daily shortwave radiation $\bar{R}_d$ [MJ m^{-2} day^{-1}]:

$$G = \frac{\bar{R}_d}{3.6} \times 365 \quad [\text{kWh m}^{-2}\text{yr}^{-1}] \tag{5}$$

PV System Hourly output for p panels of rated power P_{W} [W]:

$$P_t^{\text{PV}} = \frac{p\,P_{\text{W}}}{1000} \cdot G_t \tag{6}$$

Annual production and per-panel yield:

$$E^{\text{PV}} = \frac{p\,P_{\text{W}}}{1000} \cdot G\,\eta_{\text{PR}}, \qquad E_{\text{panel}} = \frac{P_{\text{W}}}{1000} \cdot G\,\eta_{\text{PR}} \tag{7}$$

Panel selection (no brand specified): $\text{panel}^* = \arg\max_j \frac{\eta_j}{c_j}$

Load Profile Let $\Delta\lambda = \lambda - \lambda_{\text{coast}}$ ($\lambda_{\text{coast}} = -117.25$), φ = latitude, $d_c = \sqrt{(\varphi - 32.7157)^2 + (\lambda - (-117.1611))^2}$.

$$f_1 = \begin{cases} 1.25 & \Delta\lambda \geq 0.15 \\ 1.05 + 4(\Delta\lambda - 0.10) & 0.10 \leq \Delta\lambda < 0.15 \\ 0.95 + \Delta\lambda & 0 \leq \Delta\lambda < 0.10 \\ 0.90 + (\Delta\lambda + 0.05) & -0.05 \leq \Delta\lambda < 0 \\ 0.85 & \Delta\lambda < -0.05 \end{cases} \qquad f_2 = \begin{cases} 1.10 & \varphi < 32.60 \\ 1.05 & 32.60 \leq \varphi < 32.70 \\ 1.00 & 32.70 \leq \varphi < 32.85 \\ 0.95 & 32.85 \leq \varphi < 32.95 \\ 0.90 & \varphi \geq 32.95 \end{cases} \tag{8}$$

$$f_3 = 1 + 0.15\big[\max(0, \Delta\lambda) + 2\max(0, \varphi - 32.70)\big], \quad f_4 = 0.70 + 0.10\min(n_p, 6) \tag{9}$$

$$f_5 = \begin{cases} 0.7 & d_c < 0.03 \\ 0.9 & 0.03 \leq d_c < 0.08 \\ 1.1 & 0.08 \leq d_c < 0.15 \\ 1.3 & d_c \geq 0.15 \end{cases} \tag{10}$$

Scaled load profile (f_6: coastal/urban factor; f_9: stochastic micro-grid factor):

$$L_t = L_t^{\text{base}} \cdot f_1 f_2 f_3 f_4 f_5 f_6 f_9 \tag{11}$$

EV charging overlay ($P_{\text{EV}} = 7.2\,\text{kW}$, $E_{\text{daily}} = 14.0\,\text{kWh EV}^{-1}\,\text{day}^{-1}$, hours 22:00-06:00):

$$L_t^{\text{EV}} = P_{\text{EV}}\,n_{\text{EV}} \cdot \mathbf{1}[t \bmod 24 \in \mathcal{H}_{\text{charge}}], \quad |\mathcal{H}_{\text{charge}}| = \left\lceil \frac{E_{\text{daily}}}{P_{\text{EV}}} \right\rceil \tag{12}$$

Daytime occupancy uplift ($9 \leq h < 17$):

$$L_t \leftarrow L_t \cdot (1 + 0.05\,n_{\text{day}}) \tag{13}$$

Battery Dispatch Simulation Net at hour t: $\text{net}_t = L_t - P_t^{\text{PV}}$

Surplus ($\text{net}_t < 0$):

$$c_t = \min\left(P_c^{\max},\ \frac{C - \text{SoC}_t}{\eta},\ |\text{net}_t|\right), \quad e_t = |\text{net}_t| - c_t \tag{14}$$

Deficit ($\text{net}_t \geq 0$):

$$d_t = \min(P_d^{\max},\ \text{SoC}_t,\ \text{net}_t), \quad g_t = \text{net}_t - d_t \tag{15}$$

SoC update:

$$\text{SoC}_{t+1} = \text{clip}(\text{SoC}_t + c_t\,\eta - d_t,\ 0,\ C) \tag{16}$$

Annual cost terms:

$$\text{CostImport} = \sum_t g_t\,\pi_t, \qquad \text{CreditExport} = \sum_t e_t\,\pi_{\text{NEM}} \tag{17}$$

Economic Model Capital costs ($r_{\text{inst}} = 0.10$, $r_{\text{ITC}} = 0.30$):

$$\text{CAPEX}_{\text{gross}} = \big(p\, P_{\text{W}}\, c_{\text{W}} + n_{\text{units}}\, c_{\text{bat}}\big)(1 + r_{\text{inst}}), \quad \text{CAPEX}_{\text{net}} = \text{CAPEX}_{\text{gross}}(1 - r_{\text{ITC}}) \tag{18}$$

Annual bills and year-1 savings ($F = \$0.345\,\text{day}^{-1}$, $c_{\text{O\&M}} = \$0.005\,\text{W}^{-1}\,\text{yr}^{-1}$):

$$B_1^{\text{trad}} = E^{\text{load}}\,\bar{\pi} + 365F \tag{19}$$

$$B_1^{\text{solar}} = \text{CostImport} - \text{CreditExport} + 365F \tag{20}$$

$$S_1 = B_1^{\text{trad}} - B_1^{\text{solar}} - c_{\text{O\&M}}\, p\, P_{\text{W}} \tag{21}$$

Multi-year savings with degradation rate δ and utility inflation $r_{\text{inf}} = 0.06$:

$$E_y^{\text{imp}} = E^{\text{load}} - \big(E^{\text{load}} - E_{\text{sim}}^{\text{imp}}\big)(1 - \delta)^{y-1} \tag{22}$$

$$S_y = B_1^{\text{trad}}(1 + r_{\text{inf}})^{y-1} - \Big[E_y^{\text{imp}}\,\bar{\pi}_{\text{eff}}(1 + r_{\text{inf}})^{y-1} - E_y^{\text{exp}}\,\pi_{\text{NEM}} + 365F + c_{\text{O\&M}}\, p\, P_{\text{W}}(1 + r_{\text{inf}})^{y-1}\Big] - I_y \tag{23}$$

where $E_y^{\text{exp}} = E_{\text{sim}}^{\text{exp}}(1 - \delta)^{y-1}$ and $I_{10} = \$2{,}000$ (inverter replacement), $I_y = 0$ otherwise.

NPV and payback ($r_d = 0.07$, $N = 10\,\text{yr}$):

$$\text{NPV} = -\text{CAPEX}_{\text{net}} + \sum_{y=1}^{N} \frac{S_y}{(1 + r_d)^y}, \qquad y^* = \min\left\{ y \;\middle|\; \sum_{i=1}^{y} S_i \geq \text{CAPEX}_{\text{net}} \right\} \tag{24}$$

Annuity factor and present-value multiplier (used in MILP objective):

$$\text{AF} = \frac{r_d(1 + r_d)^N}{(1 + r_d)^N - 1}, \qquad \text{PVM} = \sum_{y=1}^{N} \frac{(1 + r_{\text{inf}})^{y-1}}{(1 + r_d)^y} \tag{25}$$

MILP Formulation **Decision variables** $p \in \mathbb{Z}_{\geq 0}$ (panels); $y_k \in \{0, 1\}$ (battery option); $g_t, e_t, c_{kt}, d_{kt}, \text{SoC}_t \geq 0$.

Objective - minimise annualised life-cycle cost

$$\min \quad \underbrace{c_p^{\text{net}} \cdot p \cdot \text{AF}}_{\text{PV annuity}} + \underbrace{\sum_k c_k^{\text{bat,net}}\, y_k \cdot \text{AF}}_{\text{battery annuity}} + \underbrace{\text{PVM} \sum_t \pi_t\, g_t}_{\text{import cost}} - \underbrace{\pi_{\text{NEM}}\, \text{PVM} \sum_t e_t}_{\text{export credit}} \tag{26}$$

where $c_p^{\text{net}} = P_{\text{W}}\, c_{\text{W}}(1 + r_{\text{inst}})(1 - r_{\text{ITC}})$ and $c_k^{\text{bat,net}} = c_k^{\text{bat}}(1 + r_{\text{inst}})(1 - r_{\text{ITC}})$.

Constraints

Battery option selection:

$$\sum_k y_k = 1; \qquad C = \sum_k C_k\, y_k; \quad P^c = \sum_k P_k^{c,\max}\, y_k; \quad P^d = \sum_k P_k^{d,\max}\, y_k \tag{27}$$

Power balance $\forall\, t$:

$$\frac{p\, P_{\text{W}}}{1000} G_t + g_t - e_t + \sum_k (d_{kt} - c_{kt}) = L_t \tag{28}$$

SoC dynamics $\forall\, t$:

$$\text{SoC}_{t+1} - \text{SoC}_t + \sum_k \frac{d_{kt}}{\eta_k} - \sum_k \eta_k\, c_{kt} = 0 \tag{29}$$

SoC bounds and initialisation ($\alpha_{\min} = 0.10$, $\alpha_{\text{init}} = 0.50$):

$$\text{SoC}_0 = \alpha_{\text{init}}\, C; \qquad \alpha_{\min}\, C \leq \text{SoC}_t \leq C \quad \forall\, t \tag{30}$$

Big-M coupling (charge/discharge only when option k selected):

$$c_{kt} \leq M\, y_k, \quad d_{kt} \leq M\, y_k \quad \forall\, k, t \tag{31}$$

Export cap, roof area, and budget:

$$e_t \leq \tfrac{p\, P_{\text{W}}}{1000} G_t \quad \forall\, t \tag{32}$$

$$p\, A_{\text{panel}} \leq A_{\text{roof}}, \quad A_{\text{roof}} = \max\big(\lfloor \tfrac{W}{w} \rfloor \lfloor \tfrac{L}{\ell} \rfloor,\ \lfloor \tfrac{W}{\ell} \rfloor \lfloor \tfrac{L}{w} \rfloor\big)\, A_{\text{panel}} \tag{33}$$

$$p\, c_{\text{panel}}(1 + r_{\text{inst}}) + \sum_k c_k^{\text{bat}}(1 + r_{\text{inst}})\, y_k \leq B \tag{34}$$

String Configuration Optimal series/parallel split minimising deviation from 400 V inverter target:

$$n_s^* = \arg \min_{n_s | n_s \cdot n_p = n} |n_s V_{\text{mp}} - 400|, \quad V_{\text{string}} = n_s^* V_{\text{mp}} \tag{35}$$

Cell-level quantities:

$$V_{\text{cell}} = \frac{V_{\text{mp}}}{N_s^{\text{cell}}}, \qquad I_{\text{cell}} = \frac{P_{\text{W}}}{N_s^{\text{cell}} N_p^{\text{cell}} V_{\text{cell}}} \tag{36}$$

To ensure robustness under variability in weather and demand, the formulation is extended over scenarios:

$$\min \sum_{s \in \mathcal{S}} \pi_s \cdot C_{\text{total},s}(S_{\text{PV}}, E_{\text{BESS}}) \tag{37}$$

where π_s represents the probability of each scenario.

The model outputs system-level recommendations as optimal sizing decisions $(S_{\text{PV}}^*, E_{\text{BESS}}^*)$, with associated cost and savings estimates. In the agentic pipeline, outputs are structured into two scenario-level recommendations (*optimal* and *recommended*) plus a shared evidence block. Each scenario includes sizing parameters, energy estimates, financial metrics, constraints, assumptions, and confidence, yielding a machine-readable record. A pre-defined JSON schema specifies required fields, data types, and valid ranges to ensure consistency, interpretability, and downstream compatibility.

A recommendation is valid only if physical and logical constraints are satisfied, including feasible sizing, bounded operational states (e.g., $\text{SoC}(t)$), and time-step power-balance consistency. Validation proceeds in three stages: syntactic (valid JSON), schema (structural correctness and bounds), and logical (domain checks such as budget feasibility and realistic configurations). If any check fails, an automatic repair step prompts the model to regenerate output under the same schema constraints. This validation-repair loop improves robustness while preserving evidence traceability to computed features and retrieved context.

A.3 Feature-engineering workflow

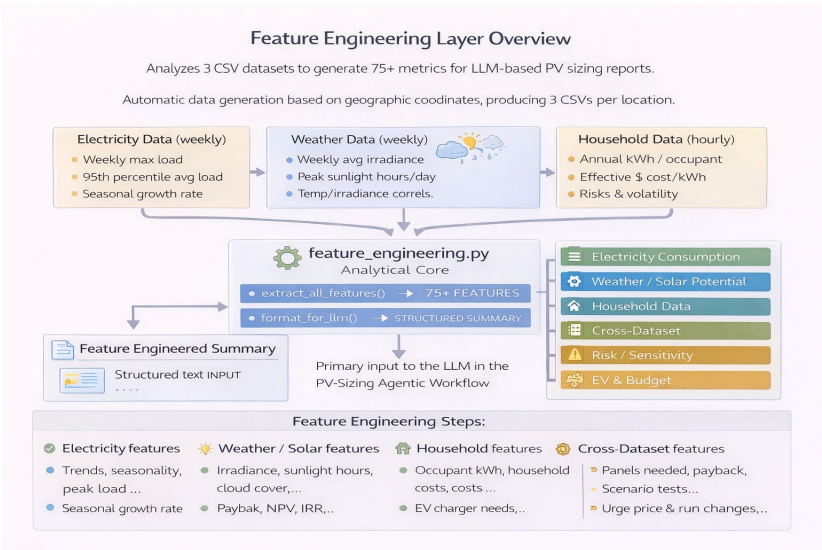


Figure 5: Feature engineering and aggregation pipeline used to transform raw weather/load signals into structured optimization-ready features.

A.4 Additional analysis plots

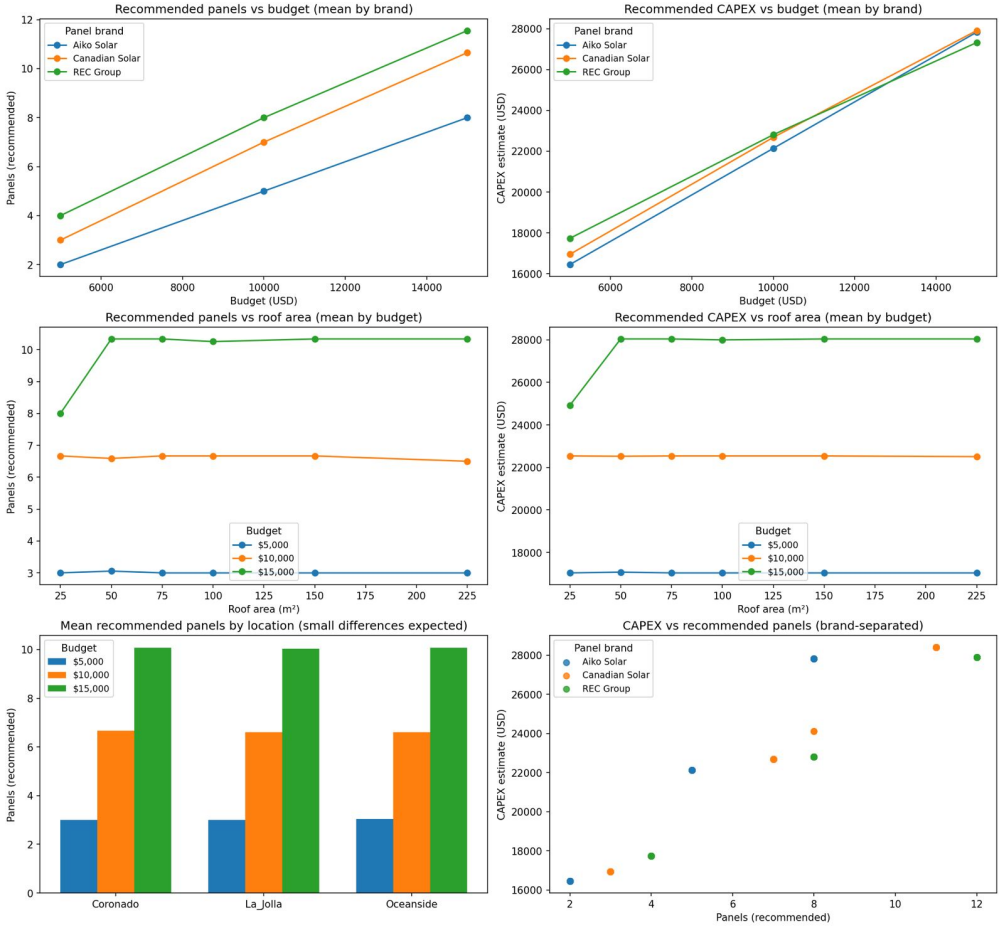


Figure 6: Analysis plot for case 243.

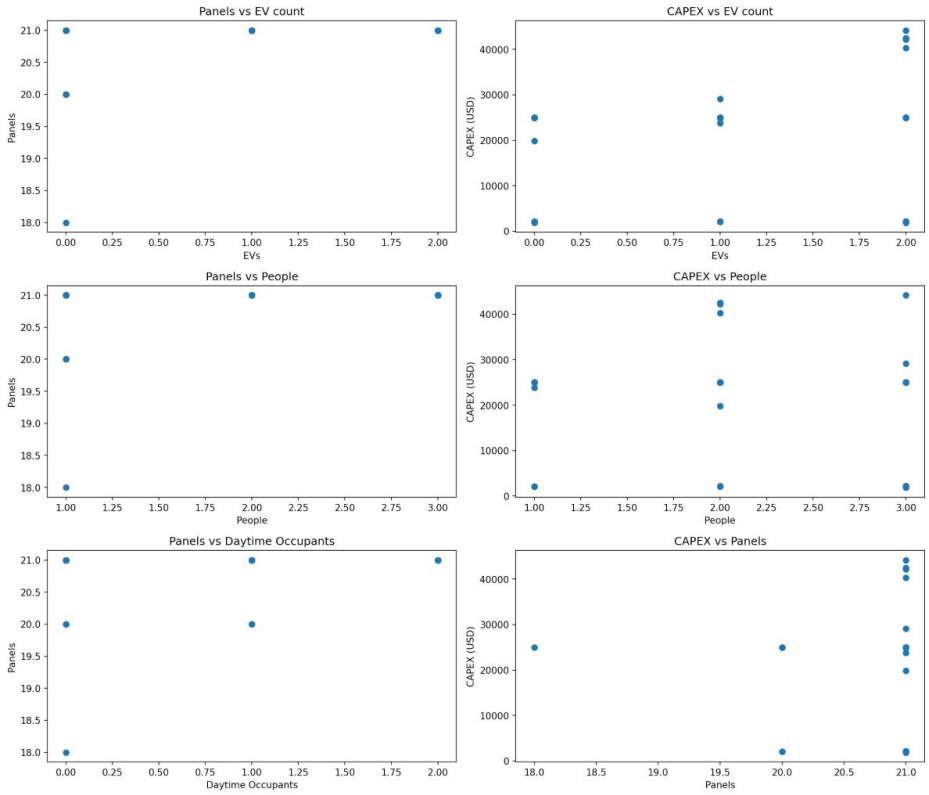


Figure 7: Analysis plot for case 72.

A.5 Budget-constrained head-to-head table

Table 9: Track A vs Tracks B1/B2 head-to-head **Budget-constrained sizing**. Error measured against Baseline reference; ✓indicates the better metric value.

Metric	Track A	Track B1	Track B2	Better
Panels RMSE	86.44	**37.80**	**37.80**	B1/B2 ✓
Panels MAE	63.70	**26.00**	**26.00**	B1/B2 ✓
Capex MAE ($)	18,220	**7,670**	**7,670**	B1/B2 ✓
Capex MAPE (%)	1,738.2	**732.2**	**732.2**	B1/B2 ✓
Feasibility (%)	**85.0**	75.0	75.0	Track A ✓
Violations	**3/20**	5/20	5/20	Track A ✓

A.6 San Diego locations

Location	Latitude	Longitude
San Diego (Downtown)	32.7160	-117.1611
Chula Vista	32.6399	-117.1067
Oceanside	33.1959	-117.3795
Escondido	33.1247	-117.0808
Carlsbad	33.1581	-117.3506
Vista	33.1936	-117.2411
San Marcos	33.1434	-117.1661
Encinitas	33.0391	-117.2954
National City	32.6692	-117.0890
Imperial Beach	32.5839	-117.1131
El Cajon	32.7948	-116.9628
La Mesa	32.7678	-117.0230
Lemon Grove	32.7335	-117.0337
Santee	32.8384	-116.9739
Poway	32.9627	-117.0362
Solana Beach	33.0006	-117.2688
Alpine	32.8351	-116.7664
Bonita	32.6653	-117.0444
Fallbrook	33.3764	-117.2511
Jamul	32.7266	-116.8823
Coronado	32.6859	-117.1831
Del Mar	32.9595	-117.2653
Rancho Santa Fe	33.0167	-117.2056
Camp Pendleton North	33.2817	-117.3197
Eucalyptus Hills	32.8295	-116.8201
La Jolla	32.8427	-117.2578
Mira Mesa	32.9185	-117.1382
Lakeside	32.8820	-116.9017
Casa de Oro-Mt. Helix	32.7335	-116.9964
Bostonia	32.8041	-116.9470

Table 10: Locations used in benchmark and grid evaluations.

A.7 TOU tariff assumptions

Period	Buy ($/kWh)	Export ($/kWh)
Off-peak	0.32	0.08
Mid-peak	0.44	0.10
On-peak	0.59	0.13

Table 11: Representative TOU tariffs used for evaluation.

A.8 Unconstrained head-to-head table

Table 12: Unconstrained head-to-head comparison of Track A and Tracks B1/B2 against the Baseline reference; ✓indicates the better metric value.

Metric	Track A	Track B1	Track B2	Better
Panels RMSE	277.35	**20.05**	**20.05**	B1/B2 ✓
Panels MAE	186.85	**14.10**	**14.10**	B1/B2 ✓
Capex MAE ($)	60,585	**26,902**	**26,902**	B1/B2 ✓
Capex MAPE (%)	10,468.5	**1,365.2**	**1,365.2**	B1/B2 ✓
Feasibility (%)	25.0	**40.0**	**40.0**	B1/B2 ✓
Violations	15/20	**12/20**	**12/20**	B1/B2 ✓

Note: B1 and B2 are numerically identical; both are shown for completeness.

A.9 Hardware and cost assumptions

Panel brand	Efficiency (%)	Cost ($/W$_p$)	Power (W)
REC Group (Alpha Pure)	21.9	0.83	405
JA Solar (DeepBlue 3.0)	20.2	0.45	395
Trina Solar (Vertex S)	20.8	0.32	400
Canadian Solar (TOPHiKu7)	23.2	0.16	720
Silfab Solar (Prime)	22.6	0.53	440
Jinko Solar (Tiger Neo)	22.5	0.45	440
LONGi Solar (Hi-MO 6)	22.3	0.30	435
Maxeon Solar (Maxeon 7)	24.1	3.50	445
Aiko Solar (Neostar 2P)	24.3	0.55	485

Table 13: Panel assumptions used in scenario studies.

Cost component	Value
Inverter and BOS adder	$0.55/W$_p$
Installation labor adder	$0.40/W$_p$
Battery pack cost	$410/kWh
Annual O&M rate	1.0% of capex

Table 14: Cost constants used in optimization and financial analysis.

Table 15: Hardware Costs and Baseline Tariff Parameters for the San Diego Model

Parameter	**Value**	**Source**
Average PV Installed Cost	\$2.85-\$3.75/W_p (avg. \$3.14/W)	EnergySage [2025]
Average BESS Installed Cost	\$7,000-\$18,000/unit (10-13.5 kWh)	Industry data (2025)
TOU On-Peak Rate (Summer 2025)	\$0.599/kWh	San Diego Gas & Electric [2025]
TOU Off-Peak Rate (Summer 2025)	\$0.528/kWh	San Diego Gas & Electric [2025]
TOU Super Off-Peak (Summer 2025)	\$0.450/kWh	San Diego Gas & Electric [2025]
Daily Fixed Charge (2025)	\$0.402/day (Min. Bill)	San Diego Gas & Electric [2025]
NEM 3.0 Export Compensation	\$0.05-\$0.08/kWh	California Public Utilities Commission [2022]
Panel Temp. Coefficient (γ)	-0.004 %/°C	Jordan and Kurtz [2013]
Performance Ratio (PR)	0.75-0.85	IEA-PVPS (2020)
Annual Degradation Rate (d)	0.5%/year	Jordan and Kurtz [2013]
BESS Round-Trip Efficiency	90-95%	Manufacturer datasheets
Annual O&M Cost	\$350-\$500/year	National Renewable Energy Laboratory [2024b]